soils.sol

Purpose

`soils.sol` defines every soil class used in the watershed. Each class has a profile header (name, number of layers, hydrologic group, rooting depth, anion exclusion fraction, crack volume, texture string) and one row per soil layer describing depth, bulk density, available water capacity, saturated hydraulic conductivity, organic carbon, particle-size fractions, rock content, albedo, USLE K, electrical conductivity, calcium carbonate, and pH. HRUs reference these classes through `hru-data.hru`. Values feed `soil_phys_init`, `soil_nutcarb_init`, runoff, percolation, erosion, and plant uptake routines.

Source

- Reader: src/soil_db_read.f90
- Type definition: src/soil_data_module.f90, `type soil_database` (which holds `type soil_profile_db` plus an allocatable array of `type soilayer_db`)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Each soil is a multi-record block:
  - One profile record with 7 fields.
  - `nly` layer records that follow, each with 14 fields. `nly` is read from the second field of the profile record and used to allocate the layer array.
- Records are list-directed (free format). Column positions in the reference example are cosmetic indentation only; the reader does not depend on them.

Profile record (7 fields)

Read into `soildb(i)%s` (`type soil_profile_db`).

#	Field	Type	Units	Default	Description
1	snam	char(20)	none	blank	soil class name. Referenced from `hru-data.hru`
2	nly	int	none	1	number of soil layers that follow this record
3	hydgrp	char(16)	none	“A”	hydrologic soil group (A, B, C, or D)
4	zmx	real	mm	1500.0	maximum rooting depth
5	anion_excl	real	fraction	0.5	fraction of porosity from which anions are excluded
6	crk	real	fraction	0.01	crack volume potential of the soil
7	texture	char(16)	none	blank	texture descriptor string (free-form label)

Layer record (14 fields, repeated `nly` times)

Read into `soildb(i)%ly(j)` (`type soilayer_db`) for `j = 1..nly`.

#	Field	Type	Units	Default	Description
1	z	real	mm	1500.0	depth from soil surface to the bottom of the layer
2	bd	real	Mg/m^3	1.3	moist bulk density
3	awc	real	mm H2O / mm soil	0.2	available water capacity
4	k	real	mm/hr	10.0	saturated hydraulic conductivity
5	cbn	real	%	2.0	organic carbon content of the layer
6	clay	real	%	10.0	clay fraction of mineral material
7	silt	real	%	60.0	silt fraction of mineral material
8	sand	real	%	30.0	sand fraction of mineral material
9	rock	real	%	0.0	rock fragment content of the layer
10	alb	real	none	0.1	moist soil albedo
11	usle_k	real	none	0.2	USLE soil erodibility (K) factor
12	ec	real	dS/m	0.0	electrical conductivity
13	cal	real	%	0.0	CaCO3 content
14	ph	real	none	0.0	soil pH

The reader applies one fix-up after reading: if the first layer is thinner than 20 mm and either the profile has only one layer or the second layer is deeper than 20 mm, the depth of layer 1 is forced to 20 mm.

Example

Excerpt from `refdata/Ames_sub1/soils.sol`. The first soil class has four layers:

```
soils.sol Ames
   name               NLY  HYD_GRP        ZMX     ANION_EXCL      CRK       TEXTURE        DEPTH         BD
soil_01                 4        B   2000.000          0.500    0.500       L- L- L- L
                                                                                            50.00      1.260
                                                                                           150.00      1.420
                                                                                           840.00      1.500
                                                                                          2000.00      1.600
```

Notes:

- The profile record `soil_01 4 B 2000.000 0.500 0.500 "L- L- L- L"` declares 4 layers. The next four physical lines are the layer records.
- Each layer record has 14 numeric values. The `TEXTURE` token on the profile line (`L- L- L- L` here) is one space-separated token only because the reader reads `texture` as a single character field; the writer pads it with hyphen-separated horizon codes.
- Header tokens in the file (`DEPTH`, `BD`, `AWC`, `K`, `CBN`, `CLAY`, `SILT`, `SAND`, `ROCK`, `ALB`, `USLE_K`, `EC`, `CAL`, `PH`) are read past as a single skipped line. The column order is fixed by the source.

Related files

- `hru-data.hru`. Each HRU points to a soil class name in `soils.sol`.
- `nutrients.sol`. Soil chemistry initialization paired with a soil class.
- `plants.plt`. Rooting depth in `soils.sol` (`zmx`) caps plant root depth.